

[illegible]

6. k -次型 $x^T A x = (x_1, \dots, x_n) \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ 仅对实对称矩阵满足性质
 $(A^T)^k = (A^k)^T = x^T A^k x = x^T A^k x$ (定理) 正定 $A \Rightarrow \forall \lambda \in \sigma(A) \Rightarrow \lambda > 0$
 特征值 $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$ 有非零解 $\Rightarrow \det(A - \lambda I) = 0$ $\text{eq}(AB) = \text{eq}(BA)$, $\text{eq}(A^T) =$
 Hadamard 不等式: A 为正定, $\det(A) \leq \prod_{i=1}^n A_{ii}$
 有列式 $A_{ij} = (a_{ij})_{i,j=1}^n \det(A_{ij}) = \det(A_{ij}) \cdot A_{ij} \cdot A_{ij} / \det(AB) = \det(A) \det(B)$
 性质 3: $a_{11} a_{22} a_{33} \dots a_{nn} = a_{11} a_{22} a_{33} \dots a_{nn} = a_{11} a_{22} a_{33} \dots a_{nn} = a_{11} a_{22} a_{33} \dots a_{nn}$
 迹 $\text{tr}(A) = \sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i$, $\text{tr}(AB) = \text{tr}(BA)$, $\text{tr}(A^T) = \text{tr}(A) \Rightarrow A = 0$ $|A^{-1}| = |A|^{-1}$
 $x^T A x = \text{tr}(A x x^T)$, $y^T x = \text{tr}(x y^T)$, $\text{tr}(A) = \sum \lambda_i$, $\|A\|_F = \sqrt{\text{tr}(A^T A)}$
 秩: 线性无关的行/列数目 满足: $m \times n$, $\text{rank}(A) \leq \min\{m, n\}$ $\Rightarrow A$ 可满秩
 定义: 独立行/列主元数 $\text{rank}(A) = r$, $r \leq \min\{m, n\} \Rightarrow A$ 可满秩 超定: $m > n$, 欠定: $m < n$
 $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$, $\text{rank}(A) = \text{rank}(A^T)$, A 为幂等 $\Rightarrow \text{rank}(A) = \text{rank}(A^2)$, $\text{rank}(A^T) = \text{rank}(A)$, $\text{rank}(A) = \text{rank}(A^T)$, A 为幂等 $\Rightarrow \text{rank}(A) = \text{rank}(A^2)$
 秩: 独立行/列主元数 $\text{rank}(A) = r$, $r \leq \min\{m, n\} \Rightarrow A$ 可满秩 超定: $m > n$, 欠定: $m < n$

7. 非奇异 A 存在 $\text{rank}(A) = n$ 行线性独立 $Ax=b$ 解在 $-\infty$ 到 $+\infty$ 任意 $(A+xy)^{-1} = A^{-1} - \frac{A^{-1}xyA^{-1}}{1 + xA^{-1}yA^{-1}}$
 $(A+UB)^{-1} = A^{-1} - A^{-1}U(B+A^{-1}U)^{-1}B^{-1}A^{-1}$ $(A-UB)^{-1} = A^{-1} + A^{-1}U(-I+A^{-1}U)^{-1}U^{-1}A^{-1}$
 $(A-UD)^{-1} = A^{-1} + A^{-1}U(-D^{-1}-U^{-1}U^{-1}D)^{-1}U^{-1}A^{-1}$
 恒 $m \geq n$ 时, 有 $\text{rank}(A) = \text{rank}(A^T A) = m$ 当 $m=n$ 时, $R = A^{-1}AA^{-1}$

8. Moore-Penrose 逆矩阵 $(A)^{\dagger} : AA^{\dagger}A = A, A^{\dagger}AA^{\dagger} = A^{\dagger}, AA^{\dagger} \text{ 与 } A^{\dagger}A \text{ 为 Hermitian}$
 计算: $A^{\dagger} = (A^H A)^{-1} A^H = A^H (A A^H)^{-1}$

9. 直积 $A \otimes B$ $\text{rank}(A \otimes B) = \text{rank}(A) \times \text{rank}(B)$ $(A \otimes I) \times (I \otimes B) = A \otimes B$
 Hadamard 积 $(A * B)_{ij} = A_{ij} B_{ij}$ Kronecker 积 $A \otimes B$ $\text{rank}(A \otimes B) = \text{rank}(A) \times \text{rank}(B)$
 $\text{vec}(A \otimes B) = \text{vec}(A) \otimes \text{vec}(B)$

10. 矩阵的行列 $\text{vec}(A) = [A_{11}, \dots, A_{1n}, \dots, A_{m1}, \dots, A_{mn}]^T$ 左 $[A \otimes B]_{\text{left}} = [A_{11}B, \dots, A_{1n}B, \dots, A_{m1}B, \dots, A_{mn}B]$
 右 $[A \otimes B]_{\text{right}} = [A_{11}B, \dots, A_{1n}B, \dots, A_{m1}B, \dots, A_{mn}B]$
 行 $\text{vec}(A) = [A_{11}, \dots, A_{1n}, \dots, A_{m1}, \dots, A_{mn}]^T$ $\text{vec}(A^T) = \text{vec}(A)^T$
 $\text{tr}(A^T B) = (\text{vec}(A)^T)^T \text{vec}(B) = \text{tr}(AB) = (\text{vec}(A)^T)^T \text{vec}(B)$
 $\text{vec}(A * B) = \text{vec}(A) \otimes \text{vec}(B) = \text{diag}(\text{vec}(A)) \text{vec}(B) / \text{vec}(B)$
 $\text{vec}(ABC) = (C^T \otimes I) \text{vec}(B)$

函数 $f(X)$	矩阵微分 $df(X)$	Jacobian 矩阵 $J_f(X)$
$X^T X$	$X^T dX + (dX^T)X$	$(I_n \otimes X^T) + (X^T \otimes I_m)K_{nm}$
XX^T	$X(dX^T) + (dX)X^T$	$(X \otimes I_n)K_{nm} + (I_m \otimes X)$
$AX^T BXC$	$A(dX^T)BXC + AX^T B(dX)C$	$((BXC)^T \otimes A)K_{nm} + C^T \otimes (AX^T B)$
$AXB^T C$	$A(dX)B^T C + AXB^T (dX^T)C$	$(BXC^T C)^T \otimes A + (C^T \otimes (AXB^T))K_{nm}$
X^{-1}	$-X^{-1}(dX)X^{-1}$	$-(X^{-T} \otimes X^{-1})$
X^A	$\sum_{j=0}^{A-1} X^{j-1}(dX)X^{A-j-1}$	$\sum_{j=0}^{A-1} (X^j)^{T-j} \otimes X^{j-1}$
$\log X$	$X^{-1}dX$	$I \otimes X^{-1}$
$\exp(X)$	$\sum_{j=0}^{\infty} \frac{1}{j!} X^j (dX) X^{j-1}$	$\sum_{j=0}^{\infty} \frac{1}{j!} (X^j)^{T-j} \otimes X^j$
迹函数 $f(X)$	微分矩阵 $df(X)$	Jacobian 矩阵 $J_f(X)/dX^T$
$\text{tr}(X)$	$\text{tr}(dX)$	I
$\text{tr}(X^{-1})$	$-\text{tr}(X^{-1}dX)$	$-X^{-1}$
$\text{tr}(AX)$	$\text{tr}(AdX)$	A
$\text{tr}(X^2)$	$2\text{tr}(X dX)$	$2X$
$\text{tr}(X^T X)$	$2\text{tr}(X^T dX)$	$2X^T$
$\text{tr}(X^A AX)$	$\text{tr}[X^A(A+A^T)dX]$	$X^A(A+A^T)$
$\text{tr}(AXX^T)$	$\text{tr}[(A+A^T)X^T dX]$	$(A+A^T)X^T$
$\text{tr}(XAX)$	$\text{tr}[(AX+XA)dX]$	$AX+XA$
$\text{tr}(AX^{-1})$	$-\text{tr}[X^{-1}AX^{-1}dX]$	$-X^{-1}AX^{-1}$
$\text{tr}(AX^{-1}B)$	$-\text{tr}[X^{-1}BAX^{-1}dX]$	$-X^{-1}BAX^{-1}$
$\text{tr}[(X+A)^{-1}]$	$-\text{tr}[(X+A)^{-1}dX]$	$-(X+A)^{-1}$
$\text{tr}(XAXB)$	$\text{tr}[(AXB+BXA)dX]$	$AXB+BXA$
$\text{tr}(XAX^T B)$	$\text{tr}[(AX^T B+B^T X^T A)dX]$	$AX^T B+B^T X^T A$
$\text{tr}(AXX^T B)$	$\text{tr}[X^T(BA+A^T B^T)dX]$	$X^T(BA+A^T B^T)$
$\text{tr}(AX^T XB)$	$\text{tr}[(BA+A^T B^T)X^T dX]$	$(BA+A^T B^T)X^T$

1. 反Hermition阵 $A = -A^*$ 实H阵 $V = V^*$ $n \times n$ n 阶
2. 置换阵: 正文 $PP^T = PP^T = I$, $P_0 A$ 对行重排, AP_0 对列重排, P^TAP 对角相同
 互换阵 $J = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $J^2 = I$, $J^T = J$
 正置换矩阵 $G =$ 置换阵 + 非零对角阵 $\rightarrow$ 直后另分解 $x(t) = A^*x(t)$
 正逆矩阵 $A^* = (A^T)^{-1}A^{-1}$, $x(t) = A^*x(t)$ 累加号后列顺序不确定
 交换阵 $K \in \text{vec}(A \otimes mn) = \text{vec}(A^T)$ 信号幅度不满足 $A = \hat{A}G$
3. 正文 $AB^T = A^T A = I$, 酉 $U^H = U^T = I$ 酉变换 $\langle x, y \rangle = \langle Ax, Ay \rangle$
 $\Leftrightarrow \|Ux\| = \|x\|$ 的列标准正交 $U^H U = I \Leftrightarrow \det(U) = 1 \Leftrightarrow A^T A = I$
 $\|Ax\| = \|x\|$ $\cos \theta = \frac{\langle Ax, y \rangle}{\|Ax\| \|y\|} = \frac{\langle x, U^H y \rangle}{\|x\| \|y\|}$
4. 上三角 U , L , 特殊值为对角元素, 特征值相同 $Bx = Ax$
5. 相似: $B = S^{-1}AS$ (S 非奇异) $B \sim A$ 特征向量线性变换 $ASx = A(Sx)$
 $|B| = |A|$, $\text{tr}(B) = \text{tr}(A)$
6. Vandermonde阵 $\begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-1} \end{bmatrix} \Big/ \begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-1} \end{bmatrix}$ $x_1 \neq x_2 \neq \dots \neq x_n$ $\det = \prod_{i=1}^n \prod_{j=1}^n (x_i - x_j)$
7. Fourier阵 $F = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & w & w^2 & \dots & w^{n-1} \\ 1 & w^2 & w^4 & \dots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \dots & w^{(n-1)^2} \end{bmatrix}$ $w = e^{-j2\pi/n}$ $F^T F = F F^T = nI$, $x = F^{-1} \hat{x} = \frac{1}{n} F^T \hat{x}$
8. Hadamard阵 H_n : n 阶元素 ± 1 且 $H_n H_n^T = H_n^T H_n = nI_n$
 (1) 1列均为1 $\rightarrow$ 规范化Hadamard阵 $n=2$ $n=4k$, Hadamard阵存在
 $\frac{1}{\sqrt{n}} H_n$ 标准正交 $\det(H_n) = n^{\frac{n}{2}}$ 构造高维正交基
 $n=2^k$ 规范化标准正交Hadamard阵 $H_n = \frac{1}{\sqrt{n}} \begin{bmatrix} H_{n/2} & H_{n/2} \\ H_{n/2} & -H_{n/2} \end{bmatrix}$ $H_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
9. Toeplitz矩阵 在角线内取值 $\begin{bmatrix} a_0 & a_1 & a_2 & \dots & a_{n-1} \\ a_1 & a_0 & a_1 & \dots & a_{n-2} \\ a_2 & a_1 & a_0 & \dots & a_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n-1} & a_{n-2} & a_{n-3} & \dots & a_0 \end{bmatrix} = [a_k]_{i,j=0}^{n-1}$ $a_i = a_{-i}^*$ 对称Toeplitz阵 $A =$
 $A = i \cdot A^* \rightarrow$ Hermitian Toeplitz $A = \begin{bmatrix} a_0 & a_1^* & \dots & a_{n-1}^* \\ a_1 & a_0 & \dots & a_{n-2}^* \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1} & a_{n-2} & \dots & a_0 \end{bmatrix}$
10. Hankel阵 $A = \begin{bmatrix} a_{00} & a_{01} & \dots & a_{0,n-1} \\ a_{10} & a_{11} & \dots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \dots & a_{n-1,n-1} \end{bmatrix}$
 (1) 2矩阵结合

1. 链式法则 $D_x f(x) = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_m} \right] D_x f(x) = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_m} \right] \frac{\partial f}{\partial x} = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_m} \right]$
 Jacobian 矩阵 $D_x f(x) = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_m} \right] \frac{\partial f}{\partial x} = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_m} \right] \frac{\partial f}{\partial x} \in \mathbb{R}^{1 \times m}$
 链式法则 $D_{vec} f(x) = \left[\frac{\partial f}{\partial vec{x}} \right] = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_m} \right] \frac{\partial f}{\partial x} = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_m} \right] \frac{\partial f}{\partial x} \in \mathbb{R}^{1 \times m}$
 链式法则 $D_{vec} f(x) = vec(D_x f(x)) = (vec(D_x f(x)))^T / D_x f(x) = \frac{vec(D_x f(x))}{(D_x f(x))^T} \in \mathbb{R}^{m \times 1}$
 2. 列编号 (梯度算子) $D_x = \frac{\partial}{\partial x} = \left[\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_m} \right]^T$
 矩阵变元梯度算子 $D_{vec} x = \frac{\partial}{\partial vec{x}} = \left[\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_m} \right]^T$
 梯度矩阵 $D_x f(x) = \frac{\partial f}{\partial x} = D_x^T f(x)$ 梯度方向或方向 $\nabla f(x) \rightarrow$ 梯度范数 $\|x\| = \|x\|_2$
 $\nabla_x f(x) = \frac{vec(D_x f(x))}{D_x f(x)} = \left(\frac{vec(D_x f(x))}{D_x f(x)} \right)^T = (D_x f(x))^T$

$$\begin{aligned} \frac{\partial f(x, y, z)}{\partial x} &= g(x) \frac{\partial f(x, y, z)}{\partial x} + f(x, y, z) \frac{\partial g(x)}{\partial x} = g(x) h(x) \frac{\partial f(x, y, z)}{\partial x} + f(x, y, z) \frac{\partial g(x)}{\partial x} \\ \frac{\partial f(x, y, z)}{\partial x} &= \frac{1}{g(x)} [g(x) \frac{\partial f(x, y, z)}{\partial x} - f(x, y, z) \frac{\partial g(x)}{\partial x}] \quad \frac{\partial g(x)}{\partial x} = \frac{dg(x)}{dy} \frac{\partial y}{\partial x} \quad D_x x = I_n \otimes I_m \\ A^T x x^T b &\rightarrow D_x f(x) = x^T (b a^T + a b^T). \quad \nabla_x f(x) = (a b^T + b a^T) x / D_x 1^T (A B) = D_x x \\ D_x (A X B) &= B^T \otimes A. \quad \nabla_x (A X B) = B \otimes A^T / D_x (A X^T B) = (B^T \otimes A) K_{nm} \quad \nabla_x (A \\ D_x (x x^T) &= (K_{nm} + I_n) (X \otimes I_m). \quad \nabla_x (x x^T) = (X^T \otimes I_m) (K_{nm} + I_n) / X^T X \\ D_x (x^T B x) &= (B X^T \otimes I_n) K_{nm} + (I_n \otimes (X^T B)). \quad \nabla_x (x^T B x) = K_{nm} (B X) \end{aligned}$$

4. $\text{tr}(dU) = d\langle \text{tr} U \rangle$, $d(UV) = dU \cdot V + U \cdot dV$, $d(X^T) = (dX)^T$, $d(A \cdot A) = dX \cdot X + X \cdot dX$, $d(F(X)) = F(X) \cdot \text{tr}(F'(X)) d(F(X))$, $d(U \otimes V) = dU \otimes V + (dU) \cdot V + U \cdot dV$, $d(\text{vec}(X)) = \text{vec}(dX)$, $d(X^{-1}) = -X^{-1} \cdot dX$

标量函数 Jacobian 排列 $d(f(x)) = \left[\frac{\partial f(x)}{\partial x_1}, \dots, \frac{\partial f(x)}{\partial x_n} \right] \begin{bmatrix} dx_1 \\ \vdots \\ dx_n \end{bmatrix} = \frac{\partial f(x)}{\partial x^T} dx = A dx$

$d(f(x)) = \text{tr}(A dx) \Rightarrow D_x f(x) = \frac{\partial f(x)}{\partial x^T} = A$, $d(f(X)) = \text{tr}(A dX) \Rightarrow D_X f(X) = \frac{\partial f(X)}{\partial X^T} = A = \nabla_X (X^T A) = (A^T + A) X$, $\text{tr}(A d(X) B) = \text{tr}(B A dX)$, $\text{tr}(A d(X) B) = \text{tr}(B A dX)$, $\nabla_X f(X) = (B \otimes A^T) + K_{nm} (D \otimes I_m)$

$d(F(X)) = A(dX) B + C(dX^T) D \Rightarrow D_X f(X) = (B^T \otimes I) + (D^T \otimes C) K_{nm}$

5. **Hessian** $H[f(x)] = \frac{\partial^2 f(x)}{\partial x \partial x^T} = \frac{\partial}{\partial x} \left[\frac{\partial f(x)}{\partial x^T} \right] = \nabla_x^2 f(x) = \nabla_x (Df(x))$

$d^2 f(x) = (dx)^T B dx \Rightarrow H f(x) = \frac{1}{2}(B^T + B)$, $d^2 f(x) = (d(\text{vec } x))^T B dx \Rightarrow H f(x) = \frac{1}{2}(U^T V + V^T U)$
 $f(x) = w(x^T A x) \Rightarrow H f(x) = I \otimes (A + A^T)$
 形式偏导 $\frac{\partial}{\partial x} = \frac{1}{2}(\frac{\partial}{\partial x} - j \frac{\partial}{\partial y})$, $\frac{\partial}{\partial x^*} = \frac{1}{2}(\frac{\partial}{\partial x} + j \frac{\partial}{\partial y})$ 实/虚部独立假设
 $\nabla_{\mathbf{z}} f(\mathbf{z}, \mathbf{z}^*) = \frac{\partial f(\mathbf{z}, \mathbf{z}^*)}{\partial \mathbf{z}} \Big|_{\mathbf{z}^* = \bar{\mathbf{z}}} = (D_{\mathbf{z}} f(\mathbf{z}, \mathbf{z}^*))^T \Big|_{\mathbf{z}^* = \bar{\mathbf{z}}} / d f(\mathbf{z}, \mathbf{z}^*) =$

$$\begin{cases} \frac{\partial \mathbf{z}^T}{\partial \mathbf{z}} = \mathbf{I}, \frac{\partial \mathbf{z}^H}{\partial \mathbf{z}} = 0, \frac{\partial \mathbf{z}}{\partial \mathbf{z}^T} = \mathbf{I}, \frac{\partial \mathbf{z}^*}{\partial \mathbf{z}^T} = 0 \\ \frac{\partial \mathbf{z}^T}{\partial \mathbf{z}^*} = 0, \frac{\partial \mathbf{z}^H}{\partial \mathbf{z}^*} = \mathbf{I}, \frac{\partial \mathbf{z}}{\partial \mathbf{z}^*} = 0, \frac{\partial \mathbf{z}^*}{\partial \mathbf{z}^*} = \mathbf{I} \end{cases}$$

$$\begin{aligned} \text{b)}: \phi(x) &= (f(x))^T A g(x) \text{ ist } D_x \phi(x) = (g(x))^T A^T D_x f(x) + (f(x))^T A D_x g(x) \\ (D_x \phi(x))_i &= \frac{\partial (f(x))^T A g(x)}{\partial x_i} = \frac{\partial (f(x))^T}{\partial x_i} A g(x) + (f(x))^T A \frac{\partial g(x)}{\partial x_i} = (g(x))^T A^T \frac{\partial f(x)}{\partial x_i} + (f(x))^T A \frac{\partial g(x)}{\partial x_i} \\ d(f(x))^T &= dx^T \frac{\partial f(x)}{\partial x}, \quad dg(x) = \frac{\partial g(x)}{\partial x} dx \\ (Ax + b)^T (Dx + c) &\rightarrow H = A^T D + D^T A \\ d|x| &= |x|^{-1/2} (x^T dx), \quad D_x |x| = |x|^{-1/2} x^T, \quad \nabla_x |x| = |x|^{-1/2} x^T \end{aligned}$$

估计误差的协方差阵 $E\{C - \hat{C}_{WLS}}\{C - \hat{C}_{WLS}\}^T$, 最小二乘估计误差 $\min_{\hat{C}} \frac{\partial \|W(C - \hat{C})\|^2}{\partial \hat{C}}$

相似化: 特征值重数函数 (代数多重度 k / 几何多重度 $n - \text{rank}(A - \lambda I)$)

实阵 / 单根 $\checkmark$ $\text{rank}(A - \lambda I) = n - k \checkmark$

等价: 依此问题 + 约束条件

$A\theta + \varepsilon = x$, ε 为加性有色噪声, $E\{\varepsilon\} = 0$, $E\{\varepsilon\varepsilon^T\} = R$ (已知)

加权误差函数 $Q(\theta) = \varepsilon^T W \varepsilon$ (W 为对称阵), $\frac{\partial Q}{\partial \theta} = 2A^T W A \theta - 2A^T W x = 0$

$\hat{C}_{WLS} = (A^T W A)^{-1} A^T W x$, $K = E\{(\hat{C} - \hat{C}_{WLS})(\hat{C} - \hat{C}_{WLS})^T\} \leq \frac{\partial \text{tr}(K)}{\partial W} = 0 \Rightarrow \text{Wopt} = R^{-1}$

方程解情况: $m \geq n$ $\text{rank}(A) = \text{rank}[A \ b] = n$, 唯一解

$\text{rank}(A) = \text{rank}[A \ b] < n$, 无穷

$\text{rank}(A) < \text{rank}[A \ b]$, 无

$$\frac{\partial g(x)}{\partial x} + f(x)g(x)\frac{\partial h}{\partial x}$$
$$\begin{aligned} \partial &= A(dx)B \\ d\omega \otimes v + u \otimes dv \\ x^i \\ \pi(A dx) \\ J_{x^i}(x) = A^T \\ (A^T B^T dx) \\ \mathbb{T} \\ \left[\frac{\partial^2}{\partial x^1 \partial x^1}, \dots, \frac{\partial^2}{\partial x^1 \partial x^n} \right] \end{aligned}$$
$$f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} \end{bmatrix} \text{ 实对称}$$

